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R13

Code No: 117FZ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, October/November - 2020
OPERATIONS RESEARCH
(Common to ME, CSE, MCT, AME, MIE, AGE)

Time: 2 hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1. The standard weight of a special purpose brick is 5 kg and it contains two basic ingredients X and Y. X costs Rs. 5 per kg and Y costs Rs. 8 per kg. Strength considerations dictate that the brick contains not more than 4 kg of X and a minimum of 2 kg of Y. Since the demand of the product is likely to be related to the price of the brick, find graphically the minimum cost of the brick satisfying the above conditions. [15]

2. Use simplex method to
Minimize $Z= 2x_1+9x_2+x_3$
subject to constraints: $x_1+4x_2+2x_3 \geq 5$
$3x_1+x_2+2x_3 \geq 4$
$x_1>0, x_2>0, x_3>0$ [15]

3. Find the basic feasible solution and find the optimal solution with cost using Vogel's approximation method? [15]

	D1	**D2**	**D3**	**D4**	**supply**
O1	1	2	-2	3	70
O2	2	4	0	1	38
O3	1	2	-2	5	32
Demand	40	28	30	32	

4. Solve the following travelling salesman problem and find the shortest possible path.[15]

	A	B	C	D	E
A	∞	3	6	2	3
B	3	∞	5	2	3
C	6	5	∞	6	4
D	2	2	6	∞	6
E	3	2	4	6	∞

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5. Find the sequence that minimizes the total time required in performing the following jobs on three machines in the order A-B-C as shown in the below table. Also, find the total elapsed time. [15]

Machine/Job	**1**	**2**	**3**	**4**	**5**	**6**
A	8	3	7	2	5	1
B	3	4	5	2	1	6
C	8	7	6	9	10	9

6. Equipment B costs Rs.10, 000. Annual operating costs are Rs.400 for the first year and then increases by Rs.800 every year. Now you have an equipment of type A which is one year old. Should you replace it with B, if so when? [15]

7. Solve the game whose payoff matrix for the player A is given in table. [15]

Player A	Player B			
		I	II	III
	I	-2	15	-2
	II	-5	-6	-4
	III	-5	20	-8

8. Minimize $y_1^2 + y_2^2 + y_3^2$ subjected to $y_1 + y_2 + y_3 = 15$ using bell. [15]

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